

Assignment 3 (20/07/2026)

Title: Effects of Sampling in Discrete Time Signals

Objectives:

1. Simulate and plot two CT signals of 10 Hz and 110 Hz for $0 < t < 0.2$ secs.
2. Sample at $F_s = 100$ Hz and plot them in discrete form.
3. Observe and note the aliasing effects.
4. Explore and learn.

Pre- Lab

Self Study: Signal, CT signals, DT signals, sampling of analog signals, sampling theorem, Nyquist rate

In Lab Tasks:

Task 1: Sampling and aliasing in Sinusoids

1. Simulate and plot two CT signals of 10 Hz and 110 Hz for $0 < t < 0.2$ secs.
 - a. Check for the correctness of the time periods of both sinusoids
2. Sample both sinusoids at $F_s = 100$ samples/sec and plot them in discrete form.
 - a. Observe the plots and explain (write) in your own words the cause and effects of what you just saw

Task 2: Sampling and aliasing in Audio Signals

You have to generate a tone in python and listen to it with command. The frequency of the tone should be 1 kHz at a sampling rate of 8 kHz and the duration should be 3 sec.

Post Lab Tasks:

Task 1:

Change the value of F which is 1 kHz initially, listen to the tones (make sure your headphones or speakers are on). You can also change the F_s to observe the effects.

Sl.	F (kHz)	F_s (kHz)	Sounds like what frequency?	Reason. Specify whether aliasing is happening or not.
1	1	8		
2	2	8		
3	3	8		
4	4	8		
5	5	8		
6	6	8		
7	7	8		
8	8	8		
9	9	8		

10	-1	8		
11	-2	8		
12	-3	8		
13	-4	8		
14	9	8		
15	25	8		

Task 2:

Consider the continuous time sinusoidal signal: $x(t) = \sin(2\pi F_0 t)$. The sampled version will be $x(n) = \sin(2\pi(F_0/F_s)n)$, where n is a set of integers and T_s is the sampling interval ($T_s = 1/F_s$).

Plot the signal $x(n)$ for $F_s = 5$ kHz and take i) $F_0 = 0.5$ kHz ii) $F_0 = 2$ kHz iii) $F_0 = 3$ kHz and iv) $F_0 = 4.5$ kHz.

Explain the similarities and differences among various plots. Also mention whether aliasing occurs or not.

Task 3:

- Generate an analog signal of $F = 5$ Hz. Plot it for $t = 1$ sec. Take $F_s = 10$ Hz. Count no. of samples using python and print it.
- Repeat above exercise for $t = 2$ sec.
- Repeat above exercise for $F_s = 20$ Hz.
- Repeat above exercise for $F = 10$ Hz.
- Repeat above exercise for $F_s = 40$ Hz.

Observe the differences in above and explain.

Task 4:

A cosine signal of 40 Hz is sampled at 30 samples per second. What will be the f_d of the corresponding discrete signal?

Deliverables:

- Printed Report addressing the **in lab tasks and post lab tasks**
- Python code for each task.

Assignment Submission Due: 25/07/2026